

Project Design Phase
Proposed Solution Template

Date	1-07-2026
Team ID	
Project Name	Nutrition Assistant
Maximum Marks	2 Marks

Proposed Solution Template:

S.No.	Parameter	Description
1	Problem Statement (Problem to be Solved)	Many individuals find it difficult to monitor their daily calorie intake and nutritional values due to the lack of simple and accessible nutrition tracking tools. Existing solutions are often complex, expensive, or do not provide an intuitive way to track food consumption and monitor overall nutritional health.
2	Idea / Solution Description	The proposed solution is a Nutrition Assistant, a MERN Stack web application that enables users to search food items, view nutritional information, track daily calorie intake, monitor macronutrients (protein, carbohydrates, and fats), and visualize their nutritional progress through an interactive dashboard. The application provides a responsive, user-friendly interface with secure authentication and cloud-based deployment for easy accessibility.
3	Novelty / Uniqueness	The application integrates food search, calorie tracking, macronutrient analysis, and nutrition visualization into a single lightweight web platform. Unlike many nutrition applications, it focuses on simplicity, responsive design, fast performance, and a modular MERN Stack architecture that supports future enhancements such as AI-based nutrition recommendations, barcode scanning, and meal planning.
4	Social Impact / Customer Satisfaction	The solution encourages healthier eating habits by helping users understand their nutritional intake and monitor their dietary progress. It promotes health awareness, supports preventive healthcare, and provides an easy-to-use platform that can improve customer satisfaction through accurate nutrition tracking and an intuitive user experience.
5	Business Model (Revenue Model)	The application can adopt a Freemium model where basic nutrition tracking is available free of charge, while premium features such as personalized diet plans, advanced analytics, meal planning, health reports, and AI-based recommendations are offered through monthly or yearly subscription plans. Additional revenue can be generated through partnerships with nutritionists, fitness centers, and health product providers.
6	Scalability of the Solution	The solution is built using the MERN Stack, following a three-tier architecture that separates the presentation, business logic, and database layers. The frontend (React) and backend (Node.js/Express) can be scaled independently, while MongoDB Atlas supports horizontal database scaling. Cloud deployment using Vercel and Render enables the application to

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		handle increasing numbers of users and nutrition records efficiently, making it suitable for future expansion and feature integration.